Name_____ Date_____ Period_____

**LESSON 9-4 · HOMEWORK PACKET · SUPPORTED · SESSION 1 · PAGES 1–2**

# Substitute and Solve

Maryland Standard 6.EE.C.9 · Learning target – I can substitute a value into an equation to find the matching output.

**REMEMBER**

- To find the output, **substitute** the input for *x*.
- Then use **order of operations**: multiply first, then add or subtract.

**WORKED EXAMPLE**

Use $y = 6x + 4$ to find $y$ when $x = 5$.

**Step 1** Substitute: $y = 6 \times 5 + 4$

**Step 2** Multiply, then add: $30 + 4 = 34$

**Answer** **$y = 34$, so (5, 34)**

**WORD BANK**

**substitute** – put the number in place of the letter

**order of operations** – multiply first, then add or subtract

**output** – the answer you get (y)

## PRACTICE

*show your work · label units*

**1.** Substitute to find *y*.

**Step 1** Put the number in place of *x*.

**Step 2** Multiply first. Then add or subtract.

**a.** $y = 4x + 7$, $x = 3$ _____

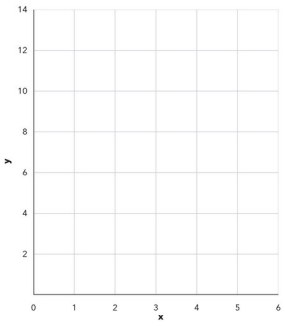
**b.** $y = 9x$, $x = 8$ _____

**c.** $y = 3x - 2$, $x = 10$ _____

**d.** $y = x + 15$, $x = 27$ _____

**2.** Complete the table for $y = 2x + 3$. Then plot the points and connect them.

| ***x*** | 0 | 1 | 2 | 4 | 5 |
|---|---|---|---|---|---|
| ***y*** | | | | | |




**3.** **Water Tank.** A tank has 20 gallons. It fills at 15 gallons each minute, so $g = 15m + 20$.

**Step 1** Put 6 in place of *m*: $15 \times 6$.

**Step 2** Then add the 20 gallons.

**a.** Gallons after 6 minutes: _____ gal

**b.** Gallons after 10 minutes: _____ gal

**4. Bike Rental.** A bike rental costs $c = 7h + 5$ dollars for $h$ hours. Find the cost for **3** hours. Then find the cost for **8** hours.

**Step 1** Multiply 7 by the hours.

**Step 2** Then add the $5.

**Costs**_____

**5. On the Line?** Substitute $x$ to decide whether each point is on the graph of $y = 5x + 2$.

**a.** (4, 22): $5 \times 4 + 2 =$ _____ On the line? ☐ Yes ☐ No

**b.** (6, 30): $5 \times 6 + 2 =$ _____ On the line? ☐ Yes ☐ No

**6. Find the Error.** For $y = 4x + 3$ with $x = 5$, Dev added first: $4 + 3 = 7$, and $7 \times 5 = 35$. Explain his mistake and find $y$.

**Frame** *Dev's mistake is ___. He should multiply first: y = ___.*

| CHECK YOURSELF | ☐ I can do this on my own | ☐ I need a little help | ☐ I have a question about #___ |
|---|---|---|---|

**AT HOME** Pick a rule like $y = 2x + 1$. Ask someone at home for an input and find the output together.

Name_____ Date_____ Period_____

**LESSON 9-4 · HOMEWORK PACKET · SUPPORTED · SESSION 2 · PAGES 3–4**

# Working Backward to a Goal

Maryland Standard 6.EE.C.9 · Learning target – I can work backward from a goal to find the input an equation needs.

**REMEMBER**

- Know the goal ($y$)? Substitute it, then solve for $x$.
- Undo in **reverse order**: subtract the fee first, then divide by the rate.
- In a counting problem, a decimal answer may need to be **rounded up**.

**WORKED EXAMPLE**

A rental costs $y = 4x + 10$ for $x$ hours. How many hours cost \$46?

**Step 1** Subtract 10: $46 - 10 = 36$

**Step 2** Divide by 4: $36 \div 4 = 9$

**Answer** **9 hours**

**WORD BANK**

**work backward** – start with the goal and undo the steps

**inverse operation** – an action that undoes another

**round up** – go to the next whole number

**goal** – the number you want to reach

## PRACTICE

*show your work · label units*

**1.** Solve for $x$.

**Step 1** Undo + or – first.

**Step 2** Then divide by the number next to $x$.

**a.** $7x = 63$ _____

**b.** $8x = 104$ _____

**c.** $3x + 5 = 32$ _____

**d.** $6x + 12 = 60$ _____

**2.** **Trip Fund.** A class has \$45 and earns \$15 a week: $y = 15x + 45$. How many weeks until it has \$210?

**Step 1** Subtract the \$45 the class already has.

**Step 2** Divide what is left by \$15 a week.

**Answer**_____ weeks

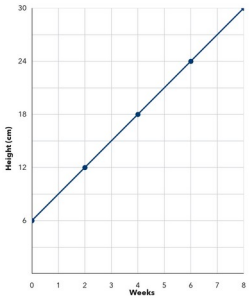
**3.** **Gift Fund.** A club has \$30 in its gift fund. Members add \$12 each week, so $y = 12x + 30$. How many weeks until the fund has \$150?

**Step 1** Subtract the \$30 already in the fund.

**Step 2** Divide what is left by \$12.

**Answer**_____ weeks

**4. Plant Growth.** A plant is 6 cm tall and grows 3 cm a week: $h = 3w + 6$.




**a.** Use the graph. When is it 18 cm? _____

**b.** Use the equation. When is it 33 cm? _____

**5. Round Up.** A box holds 12 cookies. Ana needs 150 cookies, so she says 12.5 boxes. How many boxes should she buy? Explain.

**Frame** *Ana should buy ___ boxes because ___.*

## TEST PRACTICE

*choose the best answer*

**6.** A rental costs $y = 8x + 12$ dollars for $x$ hours. Max has $60. How many hours can he rent?

**A** 5 hours **B** 6 hours **C** 7.5 hours **D** 9 hours

**7.** Mia solved $5x + 20 = 85$ by dividing 85 by 5 first. What is the correct value of $x$?

**A** $x = 17$ **B** $x = 21$ **C** $x = 13$ **D** $x = 65$

| CHECK YOURSELF | ☐ I can do this on my own | ☐ I need a little help | ☐ I have a question about #___ |
|---|---|---|---|